VCE CHEMISTRY · PRACTICE EXAM

Chemistry · Easier paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ **Date** _____

119 marks

reading 15 min · writing 150 min

State average on these questions

77 / 119 (65%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. v = sqrt(2gh) = 9.9 m/s.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
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5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the easiest questions. Questions on content that isn't in the current study design have been left out.

Sources

- 2022 Chemistry exam (PDF) · examiners' report *(previous study design)*
- 2023 Chemistry exam (PDF) · examiners' report *(previous study design)*
- 2024 Chemistry exam (PDF) · examiners' report
- 2025 Chemistry exam (PDF) · examiners' report

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–D. One mark each.

Question 1 · 1 mark

2022 VCAA exam · Section A · Question 1 · previous study design · open in the official exam PDF (p. 2) ↗

Which section of a scientific poster states the purpose of the investigation

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 2 · 1 mark

2022 VCAA exam · Section A · Question 2 · previous study design · open in the official exam PDF (p. 2) ↗

Interpreting a temperature rise in a solution calorimeter

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 3 · 1 mark

2022 VCAA exam · Section A · Question 3 · previous study design · open in the official exam PDF (p. 2) ↗

Balanced equation for incomplete combustion of ethanol

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 4 · 1 mark

2022 VCAA exam · Section A · Question 7 · previous study design · open in the official exam PDF (p. 4) ↗

Which levels of protein structure involve hydrogen bonding

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 5 · 1 mark

2022 VCAA exam · Section A · Question 8 · previous study design · open in the official exam PDF (p. 4) ↗

What a fuel cell and a galvanic cell have in common

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 6 · 1 mark

2022 VCAA exam · Section A · Question 11 · previous study design · open in the official exam PDF (p. 5) ↗

Energy profile for the uncatalysed reverse reaction

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 7 · 1 mark

2022 VCAA exam · Section A · Question 16 · previous study design · open in the official exam PDF (p. 6) ↗

IUPAC name of a chloro-alcohol from its structure

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 8 · 1 mark

2022 VCAA exam · Section A · Question 19 · previous study design · open in the official exam PDF (p. 7) ↗

Which molecule is chiral

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 9 · 1 mark

2022 VCAA exam · Section A · Question 22 · previous study design · open in the official exam PDF (p. 9) ↗

Which electrode reduction occurs at

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 10 · 1 mark

2023 VCAA exam · Section A · Question 1 · previous study design · open in the official exam PDF (p. 2) ↗

Energy change during cellular respiration

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 11 · 1 mark

2023 VCAA exam · Section A · Question 3 · previous study design · open in the official exam PDF (p. 2) ↗

Ordering fatty acids by melting point

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 12 · 1 mark

2023 VCAA exam · Section A · Question 4 · previous study design · open in the official exam PDF (p. 3) ↗

Which statement about a secondary (rechargeable) cell is correct

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 13 · 1 mark

2023 VCAA exam · Section A · Question 7 · previous study design · open in the official exam PDF (p. 3) ↗

Which statements about fuels are correct

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 14 · 1 mark

2023 VCAA exam · Section A · Question 12 · previous study design · open in the official exam PDF (p. 5) ↗

What a catalyst changes on an energy profile

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 15 · 1 mark

2023 VCAA exam · Section A · Question 21 · previous study design · open in the official exam PDF (p. 8) ↗

What denaturation does to a protein

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 16 · 1 mark

2023 VCAA exam · Section A · Question 23 · previous study design · open in the official exam PDF (p. 8) ↗

Condensed formula matching an IUPAC-named alkene

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 17 · 1 mark

2024 VCAA exam · Section A · Question 11 · open in the official exam PDF (p. 5) ↗

Naming the take–make–dispose model

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 18 · 1 mark

2024 VCAA exam · Section A · Question 13 · open in the official exam PDF (p. 6) ↗

Effect of removing a catalyst on the activation energy

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 19 · 1 mark

2024 VCAA exam · Section A · Question 30 · open in the official exam PDF (p. 11) ↗

Matching statements to poster sections (method, discussion, results)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 20 · 1 mark

2025 VCAA exam · Section A · Question 1 · open in the official exam PDF (p. 2) ↗

Renewable or not: biomethane vs natural gas

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 21 · 1 mark

2025 VCAA exam · Section A · Question 3 · open in the official exam PDF (p. 3) ↗

Why a reaction is exothermic in terms of bonds

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 22 · 1 mark

2025 VCAA exam · Section A · Question 4 · open in the official exam PDF (p. 3) ↗

Which process fits circular-economy principles

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 23 · 1 mark

2025 VCAA exam · Section A · Question 5 · open in the official exam PDF (p. 3) ↗

Calibration factor from an electrical calibration (V, I, t and ΔT given)

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 24 · 1 mark

2025 VCAA exam · Section A · Question 11 · open in the official exam PDF (p. 6) ↗

Why fuel-cell electrodes are porous

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 25 · 1 mark

2025 VCAA exam · Section A · Question 16 · open in the official exam PDF (p. 8) ↗

Effect of raising the temperature on an exothermic equilibrium

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 26 · 1 mark

2025 VCAA exam · Section A · Question 20 · open in the official exam PDF (p. 11) ↗

Type of error from judging a colour change by eye

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 27 · 1 mark

2025 VCAA exam · Section A · Question 21 · open in the official exam PDF (p. 12) ↗

Benefit of making a chemical from renewable feedstocks

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 28 · 1 mark

2025 VCAA exam · Section A · Question 22 · open in the official exam PDF (p. 12) ↗

Condensed formula of the ester propyl butanoate

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 29 · 1 mark

2025 VCAA exam · Section A · Question 24 · open in the official exam PDF (p. 12) ↗

Atom economy calculation

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Question 30 · 1 mark

2025 VCAA exam · Section A · Question 30 · open in the official exam PDF (p. 15) ↗

Predicting the ^{13}C NMR spectrum of a symmetric ketone

Your answer: ◯ **A** ◯ **B** ◯ **C** ◯ **D**

Section B – Short- and extended-answer questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (9 marks)

2023 VCAA exam · Section B · Question 3 · previous study design · open in the official exam PDF (p. 18) ↗

Combustion of propane and ethanol: energy, thermochemical equation, energy profiles, gas volume

a.i. Energy released by burning 140.0 g of propane (1 mark)

a.ii. Thermochemical equation for complete combustion of propane at SLC (2 marks)

a.iii. Sketch the energy profile (1 mark)

a.iv. How the profile for incomplete combustion differs, with justification (2 marks)

b.i. Volume of CO_2 from 96.0 g of ethanol at 15.0 °C and 100.0 kPa (3 marks)

Question 2 (7 marks)

2023 VCAA exam · Section B · Question 4 · previous study design · open in the official exam PDF (p. 20) ↗

Pathway from but-1-ene through alcohols to an acid, an amide and an ester

a.i. Type of reaction with water vapour/H_3PO_4 (1 mark)

a.ii. IUPAC name of the alcohol that cannot be oxidised to butanoic acid (1 mark)

b. Structural formula of the partial oxidation product (1 mark)

c. Skeletal formula of the product of butanoic acid and ethanamine (2 marks)

d. Atom economy for forming the ester (M = 116.0) (2 marks)

Question 3 (6 marks)

2023 VCAA exam · Section B · Question 5 · previous study design · open in the official exam PDF (p. 22) ↗

Steam reforming of methane: source of CH_4, equilibrium expression and K calculation

a. A source of methane (1 mark)

b.i. Equilibrium expression (1 mark)

b.ii. Calculate K from the starting amounts and the equilibrium amount of H_2 (4 marks)

Question 4 (16 marks)

2023 VCAA exam · Section B · Question 8 · previous study design · open in the official exam PDF (p. 30) ↗

Student investigation comparing the energy content of two types of beans by burning them under a can of water

a. Why the beans were crushed (2 marks)

b. The dependent variable (1 mark)

c. A step that manages a safety hazard, and how (2 marks)

d. Energy content of the broad bean from the results (4 marks)

e.i. Heat of combustion of soybeans from the nutrition table (2 marks)

e.ii. An error that could make the measured value (35 kJ g^{-1}) too high, and how (2 marks)

f. One improvement for more accurate results, and why (2 marks)

g. A valid conclusion (1 mark)

Question 5 (8 marks)

2024 VCAA exam · Section B · Question 2 · open in the official exam PDF (p. 14) ↗

Reaction pathway from a hydrocarbon through 1-chloropentane to an ester

a.i. Molecular formula of the starting hydrocarbon (1 mark)

a.ii. Type of reaction with Cl_2/UV (1 mark)

b. Semi-structural formula of the alcohol formed with NaOH (1 mark)

c. IUPAC name of the primary alcohol made from C_3H_6 + H_2O (1 mark)

d.i. A qualitative test for the carboxylic acid group, with the observation (2 marks)

d.ii. Skeletal structure of the ester formed (2 marks)

Question 6 (9 marks)

2024 VCAA exam · Section B · Question 3 · open in the official exam PDF (p. 16) ↗

Bread label energy; a peptide (bond, reaction, monomer); oleic vs stearic acid

a. Energy available from 100 g of bread using data-book values (1 mark)

b.i. Name of the bond joining the monomers (1 mark)

b.ii. Type of reaction forming that bond (1 mark)

b.iii. Name an amino acid in the circled section, using the data book (1 mark)

c.i. A structural difference between oleic and stearic acid (1 mark)

c.ii. Equation for complete combustion of oleic acid (2 marks)

c.iii. Predict which acid has the higher molar heat of combustion, using bond enthalpies (2 marks)

Question 7 (9 marks)

2024 VCAA exam · Section B · Question 9 · open in the official exam PDF (p. 34) ↗

Two routes to paracetamol: percentage yield, atom economy and green chemistry

a.i. % yield and % atom economy of reaction P2, compared with P1 (4 marks)

a.ii. Which reaction better fits a green chemistry principle, using the data book (2 marks)

b. Discuss both reactions against the principle of waste prevention (3 marks)

Question 8 (13 marks)

2025 VCAA exam · Section B · Question 1 · open in the official exam PDF (p. 16) ↗

Heating water with propane and bioethanol: energy, efficiency, limiting reagent in a sealed box

a. Thermochemical equation for complete combustion of propane (2 marks)

b. Energy released by 9.0 g of propane (2 marks)

c.i. Energy absorbed by the water (2 marks)

c.ii. Energy efficiency of the heating (1 mark)

d. Mass of the reactant in excess when burned in a sealed 0.125 m^3 box of air (21% O_2 at SLC) (4 marks)

e. Why a larger mass of ethanol was needed for the same temperature rise (2 marks)

Question 9 (12 marks)

2025 VCAA exam · Section B · Question 3 · open in the official exam PDF (p. 21) ↗

Methanol synthesis: reversible reactions, yield vs rate, Q vs K, equilibrium calculation

a. Define a reversible reaction (1 mark)

b.i. Effect on yield of raising the temperature (2 marks)

b.ii. One way to increase the yield at constant temperature, justified (2 marks)

c. Use Q to decide which way the system must shift (3 marks)

d. Initial amount of H_2 from the equilibrium CH_3OH concentration (4 marks)

Marking guide follows

Marking guide

Mark yourself only after you've finished. Tick the boxes you earned, enter your marks and compare with how Victoria went on each question.

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	**C**	The introduction sets up the background and the purpose. The aim is stated there, before the method.	93% correct A 5% · B 1% · C 93% · D 1%
☐	2	**A**	The water absorbs the energy the reaction releases. The temperature rises, so the reaction is exothermic and ΔH is negative.	65% correct A 65% · B 20% · C 13% · D 2%
☐	3	**C**	\(\text{C}_2\text{H}_5\text{OH} + 2\text{O}_2 \rightarrow 2\text{CO} + 3\text{H}_2\text{O}\). Balance C, then H, then O.	78% correct A 11% · B 4% · C 78% · D 8%
☐	4	**A**	Hydrogen bonds hold α-helices and β-sheets (secondary structure). They also occur between R groups (tertiary) and between subunits (quaternary). Primary structure is covalent peptide bonds only.	78% correct A 78% · B 10% · C 8% · D 4%
☐	5	**D**	In both, the oxidant and the reductant are kept apart, so electrons have to travel through the external circuit.	65% correct A 14% · B 11% · C 10% · D 65%
☐	6	**B**	The forward reaction is exothermic, so the reverse is endothermic. Uncatalysed means the higher hump. Pick the endothermic profile with the larger Ea.	63% correct A 13% · B 63% · C 7% · D 17%
☐	7	**B**	Number the chain so the –OH (the principal group) gets the lowest locant: 3-chloropentan-2-ol.	87% correct A 9% · B 87% · C 3% · D 2%
☐	8	**C**	Look for a carbon with four different groups attached. In butan-2-ol, C2 carries H, OH, CH_3 and C_2H_5.	72% correct A 13% · B 8% · C 72% · D 7%
☐	9	**D**	Reduction is always at the cathode, in galvanic and electrolytic cells alike. Only the polarity changes.	65% correct A 9% · B 22% · C 3% · D 65%
☐	10	**B**	Respiration releases energy, so it is exothermic (ΔH < 0).	67% correct A 8% · B 67% · C 8% · D 17%
☐	11	**A**	More C=C (cis) bonds put more kinks in the chain, so the chains pack less closely, the dispersion forces are weaker and the melting point is lower.	70% correct A 70% · B 2% · C 26% · D 1%
☐	12	**D**	A secondary cell works as a galvanic cell on discharge and as an electrolytic cell on recharge.	79% correct A 14% · B 2% · C 5% · D 79%
☐	13	**B**	Check each statement on its own. Only statement II survives.	73% correct A 6% · B 73% · C 21% · D 1%
☐	14	**A**	A catalyst gives an alternative pathway with a lower activation energy. ΔH and the energies of the reactants and products are unchanged.	88% correct A 88% · B 2% · C 2% · D 8%
☐	15	**D**	Denaturation disrupts the secondary and tertiary structure (hydrogen bonds and other R-group interactions). The primary sequence stays intact.	65% correct A 23% · B 7% · C 5% · D 65%
☐	16	**C**	Draw the parent chain from the name, put the double bond and the methyl branch on the numbered carbons, then check that it matches the condensed formula.	89% correct A 4% · B 4% · C 89% · D 3%
☐	17	**A**	Take, make, use and then dispose is a linear economy.	73% correct A 73% · B 18% · C 1% · D 7%
☐	18	**C**	Without the catalyst the reaction has to use the uncatalysed pathway, so the activation energy increases. ΔH does not change.	82% correct A 14% · B 2% · C 82% · D 2%
☐	19	**C**	Procedure details go in the method, interpretation goes in the discussion, and data go in the results.	81% correct A 1% · B 15% · C 81% · D 3%

☐	20	**B**	Biomethane comes from recently living biomass and can be replaced within a human timescale, so it is renewable. Natural gas is a fossil fuel, so it is not.	92% correct A 5% · B 92% · C 2% · D 1%
☐	21	**C**	Exothermic means that breaking the bonds absorbs less energy than forming the new bonds releases.	71% correct A 2% · B 5% · C 71% · D 22%
☐	22	**C**	A circular economy keeps materials in use. Turning waste CO_2 into a useful product closes the loop.	83% correct A 3% · B 9% · C 83% · D 5%
☐	23	**D**	\(CF = \frac{VIt}{\Delta T} = \frac{12.0 \times 2.50 \times 150}{6.1} = 738\ \text{J °C}^{-1}\).	93% correct A 2% · B 3% · C 2% · D 93%
☐	24	**C**	Porous electrodes give a large surface area where the gas, the electrolyte and the catalyst meet, which increases the reaction rate.	80% correct A 4% · B 4% · C 80% · D 11%
☐	25	**C**	Raising T favours the endothermic direction. For an exothermic forward reaction the position shifts left and K decreases.	67% correct A 9% · B 16% · C 67% · D 8%
☐	26	**A**	Judging an end point by eye varies unpredictably from trial to trial, so it is a random error.	64% correct A 64% · B 9% · C 17% · D 9%
☐	27	**B**	Renewable feedstocks reduce reliance on finite fossil resources. That is the green-chemistry point being tested.	84% correct A 7% · B 84% · C 6% · D 3%
☐	28	**D**	The “-anoate” part comes from the acid (4 C, including the C=O carbon) and the “propyl” part comes from the alcohol: \(\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_3\).	75% correct A 12% · B 5% · C 7% · D 75%
☐	29	**D**	Atom economy = mass of desired product ÷ total mass of reactants × 100 = 16/45 × 100 ≈ 36%.	64% correct A 9% · B 10% · C 16% · D 64%
☐	30	**A**	Pentan-3-one is symmetric, so it has 3 carbon environments, including the C=O near 210 ppm.	67% correct A 67% · B 12% · C 14% · D 7%

Multiple-choice score: ☐ **/ 30** **(state average 22.7)**

Section B – Short- and extended-answer questions

Question 1 (9 marks) · state average 5.6 / 9 (62%)

2023 VCAA exam · Section B · Question 3

a.i. Energy released by burning 140.0 g of propane

- ☐ 7.07 × 10³ kJ (with units)

Answer / notes: \(50.5 \times 140.0 = 7.07 \times 10^3\ \text{kJ}\). kJ g^{-1} as a unit lost the mark.

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

Your mark: ☐ **/ 1**

a.ii. Thermochemical equation for complete combustion of propane at SLC

- ☐ Balanced equation with correct states ($H_2O(l)$)
- ☐ ΔH = −2220 kJ mol^{-1}

Answer / notes: \(\text{C}_3\text{H}_8(g) + 5\text{O}_2(g) \rightarrow 3\text{CO}_2(g) + 4\text{H}_2\text{O}(l)\), ΔH = −2220 kJ mol^{-1}.

State average 1.3 / 2 (65%); 50% scored full marks. Mark distribution: 0: 20% · 1: 30% · 2: 50%

Your mark: ☐ **/ 2**

a.iii. Sketch the energy profile

- ☐ Exothermic profile (products below reactants, activation hump)

Answer / notes: Reactants high, a hump, and products well below the reactants.

State average 0.8 / 1 (80%); 80% scored full marks. Mark distribution: 0: 20% · 1: 80%

Your mark: ☐ **/ 1**

a.iv. How the profile for incomplete combustion differs, with justification

- [ ] Products are higher in energy than for complete combustion (still below the reactants), so less energy is released
- [ ] Because the carbon is only partly oxidised (CO/C instead of CO_2)

Answer / notes: The products sit higher (a smaller |ΔH|) because the fuel is not fully oxidised, so less energy is released. Many talked about activation energy instead.

State average 0.5 / 2 (25%); 10% scored full marks. Mark distribution: 0: 60% · 1: 40% · 2: 10%

Your mark: ☐ **/ 2**

b.i. Volume of CO_2 from 96.0 g of ethanol at 15.0 °C and 100.0 kPa

- [ ] n(ethanol) = 2.09 mol
- [ ] n(CO_2) = 4.17 mol
- [ ] V = nRT/P ≈ 99.9 L (with units)

Answer / notes: $V = \frac{4.17 \times 8.31 \times 288}{100.0} \approx 99.9\ \text{L}$. Remember to convert to kelvin.

State average 2.3 / 3 (77%); 60% scored full marks. Mark distribution: 0: 10% · 1: 10% · 2: 20% · 3: 60%

Your mark: ☐ **/ 3**

Question 2 (7 marks) · state average 4.1 / 7 (59%)

2023 VCAA exam · Section B · Question 4

a.i. Type of reaction with water vapour/H_3PO_4

- [ ] Addition (hydration)

Answer / notes: Addition (hydration).

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

Your mark: ☐ **/ 1**

a.ii. IUPAC name of the alcohol that cannot be oxidised to butanoic acid

- [ ] Butan-2-ol

Answer / notes: Butan-2-ol (the secondary alcohol).

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: ☐ **/ 1**

b. Structural formula of the partial oxidation product

- [ ] Butanal, $CH_3CH_2CH_2CHO$ drawn in full

Answer / notes: Butanal: $CH_3CH_2CH_2CHO$ with all bonds shown.

State average 0.6 / 1 (60%); 60% scored full marks. Mark distribution: 0: 40% · 1: 60%

Your mark: ☐ **/ 1**

c. Skeletal formula of the product of butanoic acid and ethanamine

- [ ] Correct amide linkage
- [ ] Correct skeletal structure (N-ethylbutanamide)

Answer / notes: N-ethylbutanamide: a 4-carbon chain ending in C(=O)–NH–, then an ethyl group.

State average 0.8 / 2 (40%); 20% scored full marks. Mark distribution: 0: 50% · 1: 30% · 2: 20%

Your mark: ☐ **/ 2**

d. Atom economy for forming the ester (M = 116.0)

- [ ] Uses atom economy = M(product)/M(all reactants) × 100
- [ ] 86.6%

Answer / notes: $\frac{116.0}{88.0 + 46.0} \times 100 = 86.6\%$.

State average 1.5 / 2 (75%); 60% scored full marks. Mark distribution: 0: 20% · 1: 20% · 2: 60%

Your mark: ☐ **/ 2**

Question 3 (6 marks) · state average 4.1 / 6 (68%)

2023 VCAA exam · Section B · Question 5

a. A source of methane

- [ ] Any valid source (natural gas, biogas, landfill, coal seam gas, anaerobic digestion)

Answer / notes: Natural gas or biogas from decomposing organic waste.

State average 0.9 / 1 (90%); 80% scored full marks. Mark distribution: 0: 20% · 1: 80%

Your mark: ____ **/ 1**

b.i. Equilibrium expression

- [ ] $K = \frac{[\text{CO}_2][\text{H}_2]^4}{[\text{CH}_4][\text{H}_2\text{O}]^2}$

Answer / notes: $K = \frac{[\text{CO}_2][\text{H}_2]^4}{[\text{CH}_4][\text{H}_2\text{O}]^2}$

State average 0.8 / 1 (80%); 80% scored full marks. Mark distribution: 0: 20% · 1: 80%

Your mark: ____ **/ 1**

b.ii. Calculate K from the starting amounts and the equilibrium amount of H_2

- [ ] Equilibrium amounts: CH_4 23.47, H_2O 21.94, CO_2 1.530, H_2 6.120 mol
- [ ] Equilibrium concentrations (÷ 100.0 L)
- [ ] Correct K value
- [ ] Correct units (M^2) and sensible sig. figs

Answer / notes: x = 6.12/4 = 1.53. Then $K = \frac{0.0153 \times 0.0612^4}{0.2347 \times 0.2194^2} = 1.90 \times 10^{-5}\ \text{M}^2$.

State average 2.4 / 4 (60%); 40% scored full marks. Mark distribution: 0: 20% · 1: 10% · 2: 10% · 3: 20% · 4: 40%

Your mark: ____ **/ 4**

Question 4 (16 marks) · state average 9.5 / 16 (59%)

2023 VCAA exam · Section B · Question 8

a. Why the beans were crushed

- [ ] Increases surface area
- [ ] So combustion is faster / more complete

Answer / notes: Crushing increases surface area, so the beans burn more completely and faster.

State average 1.5 / 2 (75%); 60% scored full marks. Mark distribution: 0: 10% · 1: 30% · 2: 60%

Your mark: ____ **/ 2**

b. The dependent variable

- [ ] Energy content (or temperature change of the water)

Answer / notes: The temperature change of the water (energy content). Just “temperature” was not enough.

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: ____ **/ 1**

c. A step that manages a safety hazard, and how

- [ ] Identifies a step and the precaution (e.g. heatproof mat, cooling water below 30 °C)
- [ ] Explains the risk it reduces (fire, burns/scalds)

Answer / notes: Step 4: the heatproof mat reduces the fire risk. Or step 8: cooling below 30 °C avoids scalds.

State average 1.8 / 2 (90%); 90% scored full marks. Mark distribution: 0: 10% · 1: 10% · 2: 90%

Your mark: ____ **/ 2**

d. Energy content of the broad bean from the results

- [ ] ΔT = 7.1 °C
- [ ] m(water) ≈ 299 g (300 mL × 0.997 g/mL)
- [ ] \(q = mc\Delta T \approx 8.88\ \text{kJ}\)
- [ ] Energy content = 8.0 kJ g^{-1} (2 sig. figs)

Answer / notes: \(299.1 \times 4.18 \times 7.1 = 8.88\ \text{kJ}\), and 8.88/1.11 = 8.0 kJ g^{-1}. Only 3% got full marks, mostly because of sig. figs.

State average 2.1 / 4 (53%); 3% scored full marks. Mark distribution: 0: 10% · 1: 20% · 2: 40% · 3: 30% · 4: 3%

Your mark: ☐ **/ 4**

e.i. Heat of combustion of soybeans from the nutrition table

- [ ] Energy per 100 g = 36×17 + 20×37 + 29×16 = 1816 kJ
- [ ] Converts to 18 kJ g^{-1}

Answer / notes: 1816 kJ per 100 g, which is 18 kJ g^{-1}. Many wrote 1816 kJ g^{-1}.

State average 1.5 / 2 (75%); 60% scored full marks. Mark distribution: 0: 20% · 1: 20% · 2: 60%

Your mark: ☐ **/ 2**

e.ii. An error that could make the measured value (35 kJ g^{-1}) too high, and how

- [ ] A plausible error that makes the result too high (e.g. mass recorded too low, thermometer touching the can, extra fuel burning)
- [ ] Explains that it makes ΔT too big or the mass too small, so ΔT/m (the energy content) is inflated

Answer / notes: The value is high, so the error must increase ΔT or decrease the recorded mass. Heat loss makes the result low, not high. 70% scored zero.

State average 0.5 / 2 (25%); 20% scored full marks. Mark distribution: 0: 70% · 1: 10% · 2: 20%

Your mark: ☐ **/ 2**

f. One improvement for more accurate results, and why

- [ ] A valid improvement (e.g. insulate, lid, shield from draughts, bomb/solution calorimeter)
- [ ] Explains how it gets the result closer to the true value (less heat lost to the surroundings)

Answer / notes: Insulate the can or add a lid so more of the released energy goes into the water, bringing the result closer to the true value.

State average 0.9 / 2 (45%); 30% scored full marks. Mark distribution: 0: 40% · 1: 40% · 2: 30%

Your mark: ☐ **/ 2**

g. A valid conclusion

- [ ] A conclusion comparing the energy content of the two beans (or noting that the soybean result is unreliable)

Answer / notes: Soybeans had a higher energy content than broad beans in this experiment.

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

Your mark: ☐ **/ 1**

Question 5 (8 marks) · state average 5.6 / 8 (70%)

2024 VCAA exam · Section B · Question 2

a.i. Molecular formula of the starting hydrocarbon

- [ ] C_5H_{12}

Answer / notes: C_5H_{12} (pentane). Keep working out of the answer box.

State average 0.8 / 1 (80%); 73% scored full marks. Mark distribution: 0: 27% · 1: 73%

Your mark: ☐ **/ 1**

a.ii. Type of reaction with Cl_2/UV

- [ ] Substitution

Answer / notes: Substitution.

State average 0.8 / 1 (80%); 75% scored full marks. Mark distribution: 0: 25% · 1: 75%

Your mark: ____ **/ 1**

b. Semi-structural formula of the alcohol formed with NaOH

- [ ] $CH_3CH_2CH_2CH_2CH_2OH$ (any valid form)

Answer / notes: $CH_3(CH_2)_4OH$. Writing "$OHCH_2$…" is an incorrect way to show a terminal –OH.

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

Your mark: ____ **/ 1**

c. IUPAC name of the primary alcohol made from $C_3H_6 + H_2O$

- [ ] Propan-1-ol

Answer / notes: Propan-1-ol.

State average 0.7 / 1 (70%); 69% scored full marks. Mark distribution: 0: 31% · 1: 69%

Your mark: ____ **/ 1**

d.i. A qualitative test for the carboxylic acid group, with the observation

- [ ] A valid test (e.g. add a metal carbonate, or pH paper/meter)
- [ ] Correct matching observation (bubbles of gas, or pH < 7 / acid colour)

Answer / notes: Add Na_2CO_3 and bubbles of CO_2 form, or test with pH paper and see pH < 7.

State average 1.2 / 2 (60%); 50% scored full marks. Mark distribution: 0: 34% · 1: 16% · 2: 50%

Your mark: ____ **/ 2**

d.ii. Skeletal structure of the ester formed

- [ ] Ester linkage drawn correctly
- [ ] Correct carbon chains on both sides (propyl pentanoate)

Answer / notes: Propyl pentanoate: a 5-carbon acid side (including C=O) joined via O to a 3-carbon propyl chain. Draw it in the box you are told to use.

State average 1.4 / 2 (70%); 60% scored full marks. Mark distribution: 0: 23% · 1: 17% · 2: 60%

Your mark: ____ **/ 2**

Question 6 (9 marks) · state average 5.9 / 9 (66%)

2024 VCAA exam · Section B · Question 3

a. Energy available from 100 g of bread using data-book values

- [ ] Correct value with units (≈ 1.0 × 10^3 kJ)

Answer / notes: \(9.5 \times 17 + 3.4 \times 37 + 47.3 \times 16 \approx 1.0 \times 10^3\ \text{kJ}\).

State average 0.7 / 1 (70%); 64% scored full marks. Mark distribution: 0: 36% · 1: 64%

Your mark: ____ **/ 1**

b.i. Name of the bond joining the monomers

- [ ] Peptide (amide) bond

Answer / notes: Peptide (amide) link.

State average 0.7 / 1 (70%); 65% scored full marks. Mark distribution: 0: 35% · 1: 65%

Your mark: ____ **/ 1**

b.ii. Type of reaction forming that bond

- [ ] Condensation

Answer / notes: Condensation.

State average 0.8 / 1 (80%); 76% scored full marks. Mark distribution: 0: 24% · 1: 76%

Your mark: ____ **/ 1**

b.iii. Name an amino acid in the circled section, using the data book

- [ ] Leucine or serine

Answer / notes: Leucine or serine. Match the side chain to the data-book table (about half the state missed this).

State average 0.5 / 1 (50%); 52% scored full marks. Mark distribution: 0: 48% · 1: 52%

Your mark: ____ **/ 1**

c.i. A structural difference between oleic and stearic acid

- [ ] Oleic acid has one C=C and stearic acid has none (or oleic has 2 fewer H)

Answer / notes: Oleic acid has one C=C double bond and stearic acid has only C–C single bonds. Say it as a direct comparison.

State average 0.7 / 1 (70%); 71% scored full marks. Mark distribution: 0: 29% · 1: 71%

Your mark: ____ **/ 1**

c.ii. Equation for complete combustion of oleic acid

- [ ] Correct reactants and products
- [ ] Correctly balanced

Answer / notes: $\text{C}_{18}\text{H}_{34}\text{O}_2 + 25\tfrac{1}{2}\text{O}_2 \rightarrow 18\text{CO}_2 + 17\text{H}_2\text{O}$

State average 1.6 / 2 (80%); 64% scored full marks. Mark distribution: 0: 10% · 1: 26% · 2: 64%

Your mark: ____ **/ 2**

c.iii. Predict which acid has the higher molar heat of combustion, using bond enthalpies

- [ ] Stearic acid, comparing C=C (614) with C–C (346) using data-book values
- [ ] Also accounts for the extra C–H bonds / extra H_2O formed (more O–H made)

Answer / notes: Stearic acid. It has C–C plus two more C–H in place of one C=C, and it forms one more H_2O on combustion, so more energy is released overall. Most students stopped at the C=C vs C–C comparison.

State average 0.9 / 2 (45%); 9% scored full marks. Mark distribution: 0: 25% · 1: 66% · 2: 9%

Your mark: ____ **/ 2**

Question 7 (9 marks) · state average 5.3 / 9 (59%)

2024 VCAA exam · Section B · Question 9

a.i. % yield and % atom economy of reaction P2, compared with P1

- [ ] % yield P2 = 41%
- [ ] % atom economy P2 = 89%
- [ ] Comparison of yields (P1 higher: 87% vs 41%)
- [ ] Comparison of atom economy (P2 higher: 89% vs 72%)

Answer / notes: Theoretical mass = $\frac{1.4}{109} \times 151 = 1.94\ \text{g}$, so yield = 0.80/1.94 = 41%. Atom economy = 151/(109 + 60) = 89%.

State average 2.6 / 4 (65%); 39% scored full marks. Mark distribution: 0: 15% · 1: 9% · 2: 18% · 3: 19% · 4: 39%

Your mark: ____ **/ 4**

a.ii. Which reaction better fits a green chemistry principle, using the data book

- [ ] Identifies atom economy as the relevant green principle (yield is not one)
- [ ] P2 is better aligned because its atom economy is higher

Answer / notes: Atom economy is the green principle, so P2 is better (89% vs 72%). Arguing from yield missed the point.

State average 1.2 / 2 (60%); 51% scored full marks. Mark distribution: 0: 31% · 1: 18% · 2: 51%

Your mark: ☐ **/ 2**

b. Discuss both reactions against the principle of waste prevention

- [ ] P1 produces waste ethanoic acid
- [ ] P2 produces waste water
- [ ] A valid comment linking yield to waste (e.g. P2's low yield means more unreacted/waste material)

Answer / notes: P1 wastes ethanoic acid, and P2 wastes only water but its low yield leaves much more unreacted material.

State average 1.5 / 3 (50%); 23% scored full marks. Mark distribution: 0: 26% · 1: 23% · 2: 28% · 3: 23%

Your mark: ☐ **/ 3**

Question 8 (13 marks) · state average 7.6 / 13 (58%)

2025 VCAA exam · Section B · Question 1

a. Thermochemical equation for complete combustion of propane

- [ ] Correctly balanced equation
- [ ] ΔH = −2220 kJ with correct SLC states

Answer / notes: $\text{C}_3\text{H}_8(g) + 5\text{O}_2(g) \rightarrow 3\text{CO}_2(g) + 4\text{H}_2\text{O}(l)$, ΔH = −2220 kJ.

State average 1.4 / 2 (70%); 42% scored full marks. Mark distribution: 0: 9% · 1: 48% · 2: 42%

Your mark: ☐ **/ 2**

b. Energy released by 9.0 g of propane

- [ ] $n(C_3H_8) = 0.2045$ mol
- [ ] Energy = 454 kJ (4.5×10^2 kJ) with units

Answer / notes: 0.2045 × 2220 = 454 kJ. Don't write "−454 kJ released".

State average 1.7 / 2 (85%); 73% scored full marks. Mark distribution: 0: 7% · 1: 20% · 2: 73%

Your mark: ☐ **/ 2**

c.i. Energy absorbed by the water

- [ ] ΔT = 66.0 °C
- [ ] q = 45.8 kJ with units

Answer / notes: $166.0 \times 4.18 \times 66.0 = 45.8\ \text{kJ}$.

State average 1.6 / 2 (80%); 68% scored full marks. Mark distribution: 0: 10% · 1: 22% · 2: 68%

Your mark: ☐ **/ 2**

c.ii. Energy efficiency of the heating

- [ ] ≈ 10%

Answer / notes: 45.8/454 × 100 ≈ 10%.

State average 0.7 / 1 (70%); 65% scored full marks. Mark distribution: 0: 35% · 1: 65%

Your mark: ☐ **/ 1**

d. Mass of the reactant in excess when burned in a sealed 0.125 m^3 box of air (21% O_2 at SLC)

- [ ] $n(O_2)$ available = 0.21 × 125 L ÷ 24.8 = 1.06 mol
- [ ] $n(O_2)$ needed = 5 × 0.2045 = 1.02 mol, so O_2 is in excess
- [ ] $n(O_2)$ left over ≈ 0.036 mol
- [ ] Mass of excess O_2 ≈ 1.1 g

Answer / notes: Remember 0.125 m^3 = 125 L. O_2 is in excess by about 0.036 mol, which is about 1.1 g.

State average 1.6 / 4 (40%); 12% scored full marks. Mark distribution: 0: 34% · 1: 19% · 2: 16% · 3: 19% · 4: 12%

Your mark: ☐ **/ 4**

e. Why a larger mass of ethanol was needed for the same temperature rise

- [ ] Ethanol has a lower energy content per gram (29.7 vs 50.5 kJ g^{-1})
- [ ] Because ethanol is already partially oxidised (it contains O)

Answer / notes: Ethanol releases less energy per gram because it is already partly oxidised. Half the state scored zero.

State average 0.6 / 2 (30%); 8% scored full marks. Mark distribution: 0: 50% · 1: 42% · 2: 8%

Your mark: ☐ **/ 2**

Question 9 (12 marks) · state average 6.9 / 12 (58%)

2025 VCAA exam · Section B · Question 3

a. Define a reversible reaction

- [ ] A reaction that can proceed in both the forward and reverse directions

Answer / notes: A reaction that can go in either direction. Defining equilibrium (equal rates) scored zero.

State average 0.9 / 1 (90%); 88% scored full marks. Mark distribution: 0: 12% · 1: 88%

Your mark: ☐ **/ 1**

b.i. Effect on yield of raising the temperature

- [ ] The forward reaction is exothermic, so the equilibrium shifts left (reverse)
- [ ] So the yield of methanol decreases

Answer / notes: Exothermic forward, so a higher T shifts it left and the yield drops, even though the rate rises.

State average 1.3 / 2 (65%); 56% scored full marks. Mark distribution: 0: 30% · 1: 14% · 2: 56%

Your mark: ☐ **/ 2**

b.ii. One way to increase the yield at constant temperature, justified

- [ ] A valid change (increase pressure by decreasing volume, add a reactant, or remove methanol)
- [ ] Justification: the system partially opposes the change by shifting forward (e.g. fewer gas particles on the right)

Answer / notes: Decrease the volume. The system shifts forward to reduce the number of gas particles (3 → 1).

State average 1.2 / 2 (60%); 37% scored full marks. Mark distribution: 0: 23% · 1: 39% · 2: 37%

Your mark: ☐ **/ 2**

c. Use Q to decide which way the system must shift

- [ ] $Q = 5.60/(2.00 \times 1.25^2) = 1.79\ M^{-2}$
- [ ] Q > K
- [ ] So the system shifts left (reverse) to return to equilibrium

Answer / notes: Q = 1.79 > K = 1.21, so the reverse reaction is favoured until Q = K.

State average 1.6 / 3 (53%); 12% scored full marks. Mark distribution: 0: 16% · 1: 19% · 2: 53% · 3: 12%

Your mark: ☐ **/ 3**

d. Initial amount of H_2 from the equilibrium CH_3OH concentration

- [ ] x = 1.30 M, so [CO] = 3.70 M at equilibrium
- [ ] \([\text{H}_2]^2 = \frac{1.30}{3.70 \times 1.21}\)
- [ ] \([\text{H}_2]_{eq} = 0.539\ \text{M}\)
- [ ] Initial H_2 = 0.539 + 2(1.30) = 3.14 mol

Answer / notes: Use an ICE table: $[H_2]eq$ = 0.539 M, and 2x = 2.60 was used up, so 3.14 mol was added.

State average 1.9 / 4 (48%); 28% scored full marks. Mark distribution: 0: 33% · 1: 14% · 2: 10% · 3: 14% · 4: 28%

Your mark: ____ **/ 4**

Your result

Total: _____ **/ 119** **State average on these questions: 77 / 119 (65%)**

Topics to revise:

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